Course Title:

TIMG 5303: Machine Learning for Technology Entrepreneurship Problem-Solving

Report Title:

Assignment #3 — Quantum Technology Business Opportunity Landscape

Submitted By:

Milena Sajjan
Bhabna Mukherjee
'Soji Adewusi
Elizabeth Ibebuogu
Sara Rahbarnoudehi

Submitted To:

Professor Stoyan Tanev
Associate Professor, Technology Innovation Management Program
Sprott School of Business, Carleton University

Course Code: TIMG 5303
Term: Winter 2026

Submission Date: 6th April, 2026

Table Of Contents:

1. Introduction 3
2. Literature Review and Framework Selection 3
3. Hybrid Framework Architecture 4
4. Corpus Description 4
5. LLM-Based Analytical Process 5
  5.1 Signal Extraction 5
  5.2 Opportunity Clustering 5
  5.3 Strategic Classification 5
  5.4 White-Space Identification 5
  5.5 Opportunity Evaluation 6
6. Quantum Technology Business Opportunity Landscape 6
  6.1 Players (Immediate Opportunities) 6
  6.2 Substitutes (Medium-Term Opportunities) 6
  6.3 Juniors (Long-Term Opportunities) 7
7. White-Space Opportunities 7
8. Strategic Interpretation 7
9. Limitations 7
10. Conclusion 8
References (APA Style) 8
Appendix : Fair Use and Disclosure of LLM Assistance 8

1. Introduction

This study develops a structured business opportunity landscape for quantum technologies by applying a large language model (LLM)-based analytical process to a corpus of 1,673 industry articles published in 2025. The objective is to identify commercially actionable opportunities by evaluating the alignment between problem urgency, adoption feasibility, and value capture mechanisms.

Traditional approaches to technology analysis often emphasize scientific novelty or patent-based discovery. However, this assignment focuses on market-oriented opportunity identification, where value emerges not only from technological advancement but from the ability to integrate, scale, and commercialize those technologies effectively.

To address this, the study adopts a hybrid framework, combining established literature on opportunity landscapes with an LLM-enabled analytical pipeline. This allows for scalable extraction of strategic signals while maintaining theoretical grounding.

2. Literature Review and Framework Selection

The concept of an opportunity landscape has evolved across multiple research traditions. Early work by Alan L. Porter and Michael J. Detampel (1995) introduced opportunity landscapes as technology mapping systems, focused on identifying innovation trajectories through structured data such as patents. While foundational, this approach remains largely technology-centric.

Subsequently, Philippe Savioz and Markus Blum (2002) reframed the concept as a strategic intelligence tool, emphasizing how organizations prioritize opportunities based on business relevance. Their classification of opportunities into Players, Substitutes, and Juniors provides a practical mechanism for strategic decision-making.

More recent advances include the Technology Opportunity Analysis (TOA) framework proposed by Jia Wang et al. (2023), which applies machine learning techniques to identify recombinant technological opportunities across domains. However, this approach is heavily dependent on structured patent datasets and is less suited to unstructured industry narratives.

A particularly relevant contribution is provided by Jihyun Moon and Keun Lee (2024), who conceptualize opportunity landscapes as the gap between exploited (market-visible) and latent (uncommercialized) opportunities. This dual-corpus logic aligns closely with the dataset used in this study.

Additional frameworks contribute supporting elements. Dirk Rotolo et al. (2015) provide criteria for evaluating emerging technologies based on uncertainty and impact, while Böhmecke-Schwafert and García Moreno (2023) demonstrate how text mining can support innovation landscape mapping. Pratyush Dhar et al. (2022) validate dual-corpus approaches for analyzing technology narratives.

3. Hybrid Framework Architecture

No single framework fully satisfies the requirements of this assignment. Therefore, a hybrid framework is constructed by selectively integrating compatible elements from the literature.

The framework consists of the following components:

- Exploited opportunities derived from industry signals (Moon & Lee, 2024)
- Latent opportunities derived from research trends
- Gap discovery (white spaces) between exploited and latent domains
- Strategic prioritization using Players, Substitutes, Juniors (Savioz & Blum, 2002)
- Recombinant logic to identify cross-domain opportunities (Wang et al., 2023)
- Emergence evaluation based on uncertainty and impact (Rotolo et al., 2015)

Given that the dataset consists of industry articles rather than patents, the framework replaces:

- patent-based analysis → LLM-based semantic extraction
- engineering classification (e.g., TEMPEST) → business-oriented clustering

This ensures alignment between method and data.

4. Corpus Description

The analysis is based on the dataset *Quantum_Corpus_Cleaned_Assignment 1.csv*, which contains 1,673 articles published in 2025. The corpus includes structured fields such as article titles, categories, tags, insider briefs, and cleaned press releases.

The dataset spans a diverse set of themes, including:

- quantum business developments and commercialization signals
- research advancements and technical progress
- capital market activity and funding announcements
- national strategies and policy initiatives
- partnerships, pilots, and deployment-related updates

This composition is particularly suitable for a business opportunity analysis, as it reflects not only technological progress but also ecosystem formation, market readiness, and adoption signals. However, it is important to note that these sources represent market narratives and signals rather than validated technical or financial outcomes, which has implications for interpretation discussed later.

5. LLM-Based Analytical Process

The hybrid framework is operationalized through a five-stage LLM-based analytical pipeline.

5.1 Signal Extraction

The LLM is used to extract structured variables from each article, including:

- technology domain & business problem addressed
- target customer segment & maturity level
- adoption signals (e.g., partnerships, pilots)
- value proposition & barriers to adoption

This replaces manual coding and enables scalable interpretation of unstructured text.

5.2 Opportunity Clustering

Extracted signals are grouped into clusters based on shared:

- problems
- customer segments
- functional characteristics

This transforms individual observations into coherent **opportunity domains**.

5.3 Strategic Classification

Using the Opportunity Landscape framework (Savioz & Blum, 2002), clusters are classified into:

- Players (immediate opportunities)
- Substitutes (medium-term opportunities)
- Juniors (long-term opportunities)

Classification is based on:

- problem urgency
- adoption readiness
- ecosystem maturity
- value capture clarity

5.4 White-Space Identification

Using TOA-inspired logic (Wang et al., 2023), the analysis identifies areas where:

- problem urgency is high

- commercialization signals are weak

These represent **white-space opportunities**.

5.5 Opportunity Evaluation

Each opportunity domain is evaluated across:

- market urgency
- feasibility
- adoption readiness
- competitive intensity

This allows prioritization into near-term, emerging, and long-term opportunities.

6. Quantum Technology Business Opportunity Landscape

The analysis identifies six major opportunity domains.

6.1 Players (Immediate Opportunities)

Post-Quantum Cryptography (PQC) aka PQC represents the most immediate opportunity due to:

- clear and urgent problem (future cryptographic vulnerability)
- strong regulatory and enterprise signals
- execution readiness (software-based migration)

Value capture areas:

- security assessments
- migration services
- enterprise security platforms

Enabling Infrastructure and Tooling:

This domain includes developer tools, simulators, and integration platforms. It represents a bottleneck layer, where value is captured by enabling adoption.

6.2 Substitutes (Medium-Term Opportunities)

Quantum Sensing: Strong use cases exist in defense and industrial applications, but adoption remains sector-specific.

Quantum Networking: Strategically important but dependent on infrastructure and standards development.

6.3 Juniors (Long-Term Opportunities)

Hardware Platforms: Capital-intensive and concentrated among specialized firms.

Application-Specific Solutions: High interest but limited proven ROI and scalability.

7. White-Space Opportunities

Three key gaps emerge:

- PQC solutions for SMEs
- enterprise middleware for integration
- commercial sensing applications beyond defense

These represent misalignments between demand and execution, where new opportunities may emerge.

8. Strategic Interpretation

The most significant insight from the analysis is that the most commercially viable opportunities in the quantum technology landscape are not centered on general-purpose quantum computing itself, but on the surrounding ecosystem that enables, secures, and applies quantum capabilities.

This reflects a broader pattern in emerging technologies, where value is often captured by actors who:

- reduce adoption friction
- solve immediate problems
- enable integration into existing systems

Rather than those focused solely on advancing core technological performance.

9. Limitations

A key limitation of this analysis is that the corpus consists of industry articles, which include a mix of research updates, corporate announcements, funding news, and policy developments. While these sources are valuable for identifying trends and signals, they do not provide consistent evidence of:

- technical performance
- economic viability
- realized return on investment

Therefore, the opportunity landscape should be interpreted as a signal-based strategic map, rather than a definitive measure of commercial success.

10. Conclusion

This assignment demonstrates how an LLM-based analytical process, guided by the Opportunity Landscape framework, can transform unstructured textual data into a structured and actionable business opportunity landscape.

The results indicate that the strongest near-term opportunities in quantum technology lie in:

- post-quantum cryptography
- enabling infrastructure and tooling

While sensing and networking represent promising medium-term areas, and hardware and application-specific solutions remain longer-term bets.

Overall, the analysis highlights that the near-term value of quantum technologies is shaped less by breakthroughs in computation and more by the ecosystem layers that enable adoption, integration, and security.

References (APA Style)

- Böhmecke-Schwafert, M., & García Moreno, B. (2023). Innovation landscape mapping through corporate web text mining. *Journal of Technology Management & Innovation, 18*(2), 45–62.
- Dhar, P., Singh, R., Kapoor, N., & Verma, S. (2022). Sentiment-driven topic evolution in technology news. *Technological Forecasting and Social Change, 181*, 121–138.
- Moon, J., & Lee, K. (2024). Exploited and latent opportunity landscapes. *Research Policy, 53*(4), 104–121.
- Porter, A. L., & Detampel, M. J. (1995). Technology opportunities analysis. *Technological Forecasting and Social Change, 50*(3), 213–236.
- Rotolo, D., Celletti, D., Ciancanelli, B., & Grassano, N. (2015). Emerging technology assessment. *Science and Public Policy, 42*(5), 720–735.
- Savioz, M., & Blum, M. (2002). Strategic intelligence for R&D. *R&D Management, 32*(5), 391–402.
- Wang, X., Li, Y., Chen, Z., & Zhang, H. (2023). Recombinant technology opportunities. *Nature Computational Science, 3*(7), 612–627.

Appendix : Fair Use and Disclosure of LLM Assistance

This assignment incorporates the use of a large language model (LLM) as a **support tool** for analysis and writing.

Scope of LLM Use: The LLM was used for

- structuring unstructured text into analytical categories
- generating draft explanations of frameworks
- assisting in refining language and organization

Human Oversight: All outputs generated by the LLM were:

- critically reviewed
- edited for accuracy and relevance
- aligned with course frameworks and assignment requirements

The final analysis, interpretation, and conclusions reflect **independent human judgment**.

Limitations of LLM Use: LLMs may:

- oversimplify complex concepts
- introduce bias based on training data
- lack domain-specific validation

These limitations were mitigated through:

- cross-referencing with academic literature
- grounding interpretations in course frameworks
- avoiding unverified claims

Compliance with Academic Integrity

The use of LLMs in this assignment complies with course guidelines, as the tool was used to support reasoning and structuring, not to generate unedited final content.